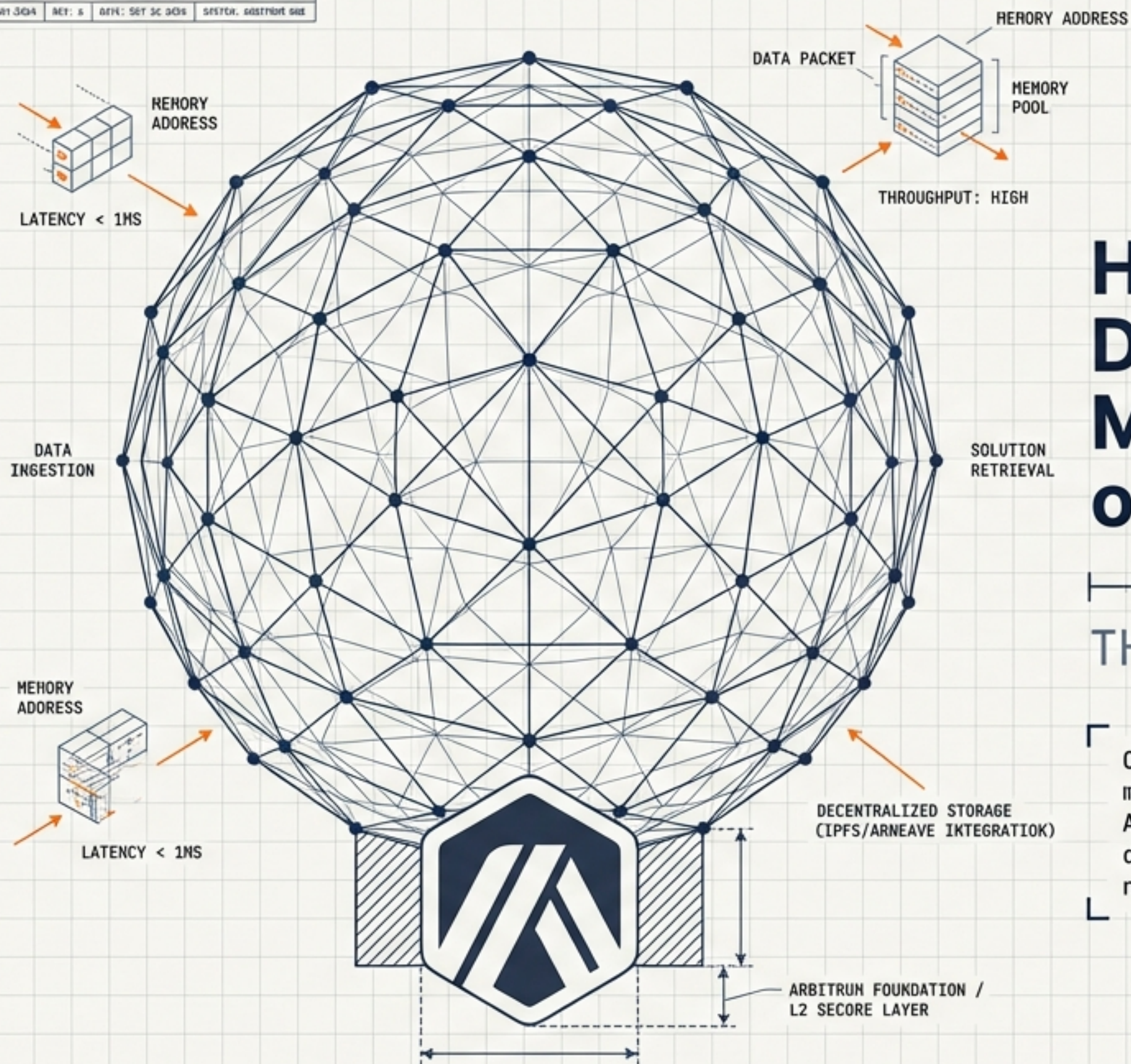
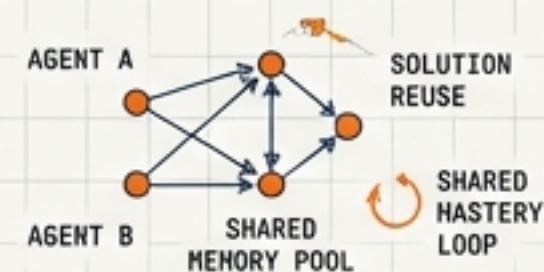




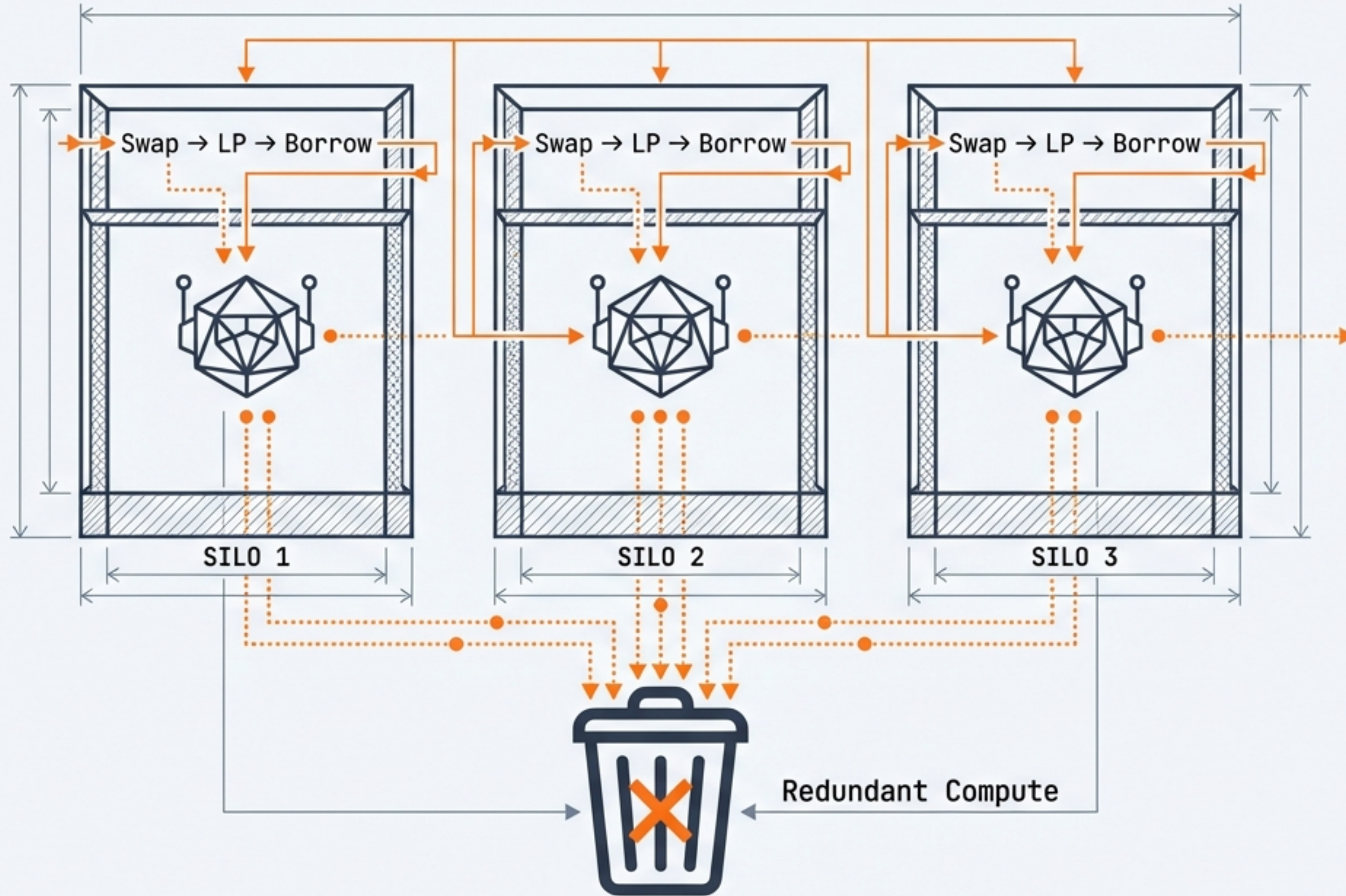
HAMN Protocol: Decentralized Associative Memory for AI Agents on Arbitrum.

THE SHARED INTELLIGENCE LAYER.

CORE CONCEPT: HAMN does not train models. It allows dApps and Agents to reuse previously discovered solutions, creating a network of shared mastery.



The Problem: The Era of Amnesiac AI



CURRENT STATE ANALYSIS:

- 1. Redundant Computation**
Re-calculating the same route thousands of times.
- 2. High Latency**
Waiting seconds for solutions that should be instant.
- 3. Isolation**
No mechanism for agents to share lessons learned.

The Core Insight

~~Input + Compute = Output~~

(Slow, Expensive)

Input + **Memory** = Instant Action

(Fast, Cheap)

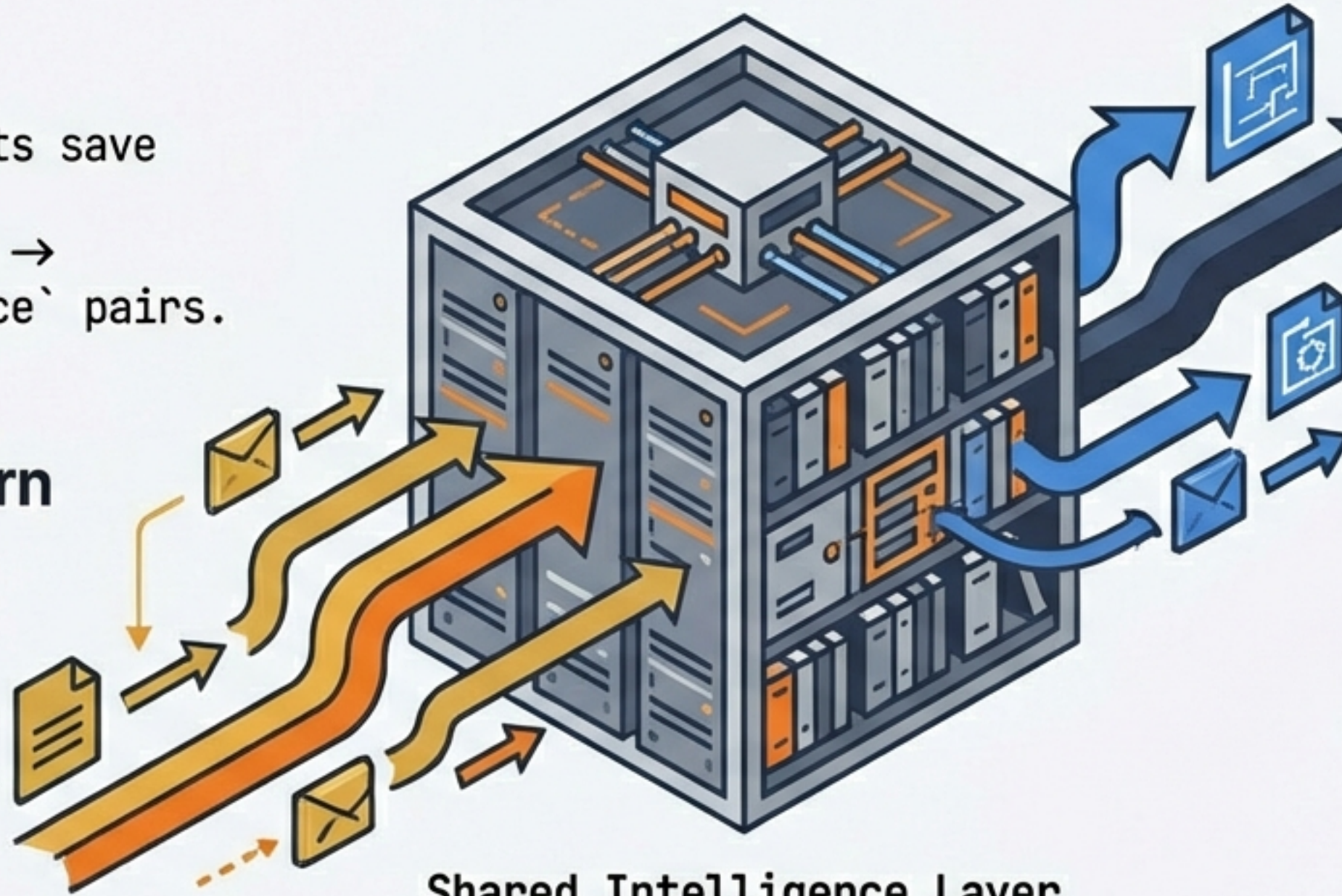
True efficiency isn't faster calculation; it's instant recall.
If a market state has been solved once, it should never be solved again.
HAMN changes the fundamental equation of AI operations.

A Marketplace for Solved Patterns

MECHANICS:

- **DEPOSIT:** Agents save successful `'market_state' → 'action_sequence'` pairs.

**Deposit Pattern
(Gold)**



Shared Intelligence Layer

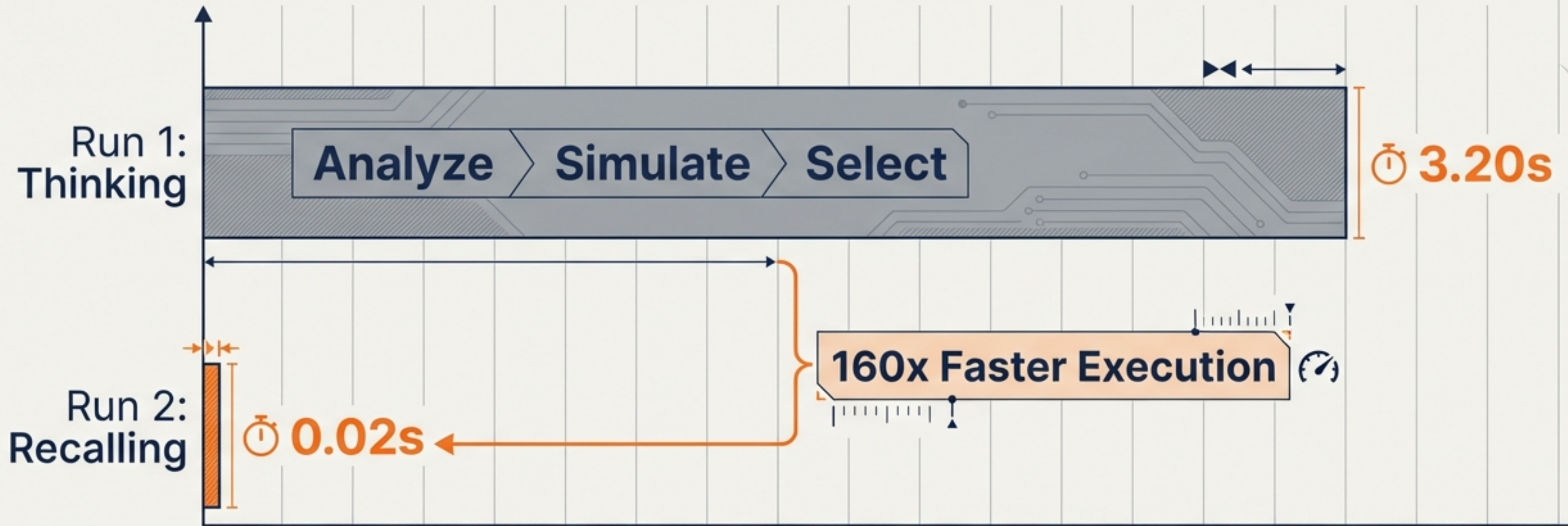
**Withdraw
Blueprint
(Blue)**

- **WITHDRAW:** Agents query the network for similar.
- **REWARD:** Contributors earn yield on pattern reuse.

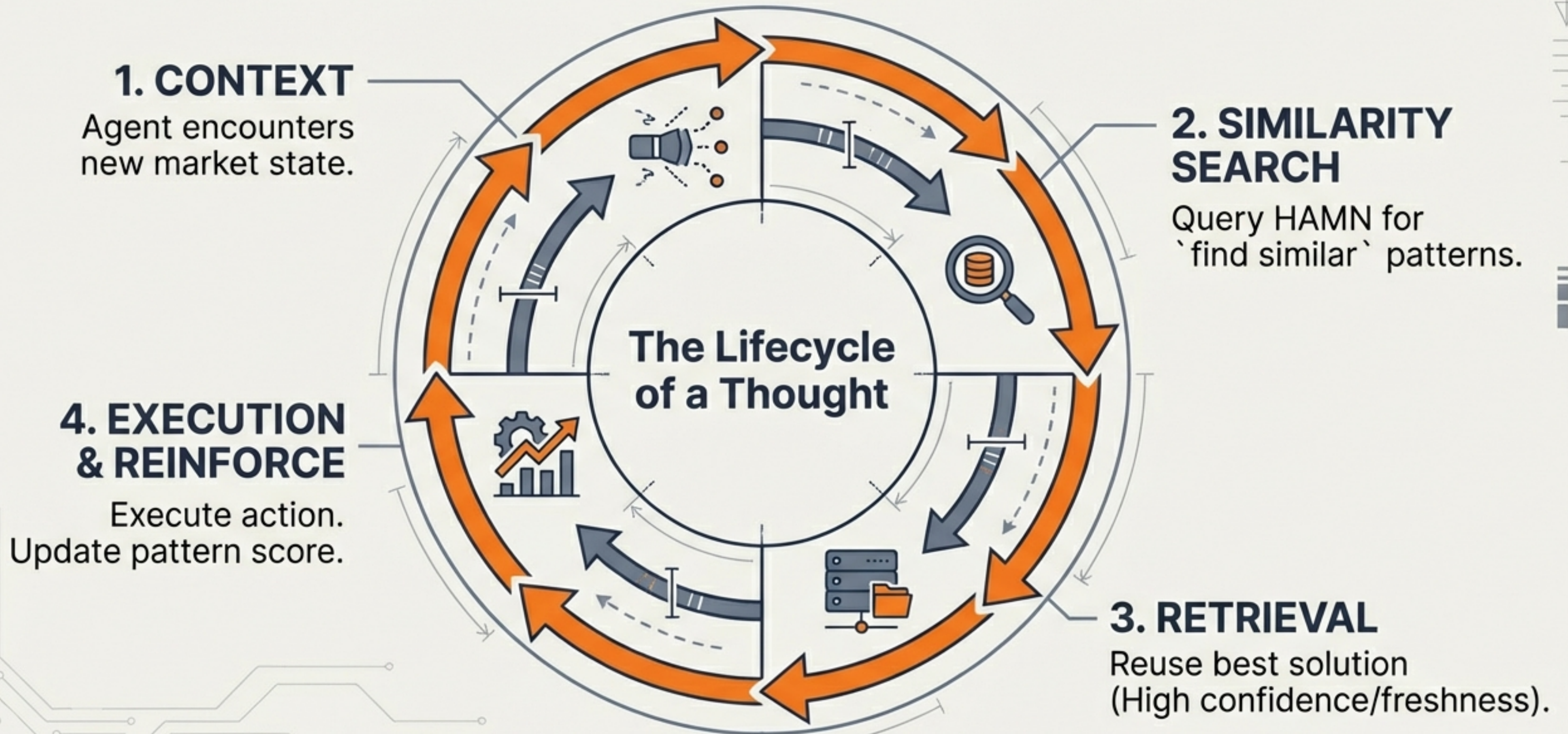
HAMN is a decentralized network where agents store successful patterns and retrieve them instantly. Contribution creates ownership.

The Speed of Memory: 160x Acceleration

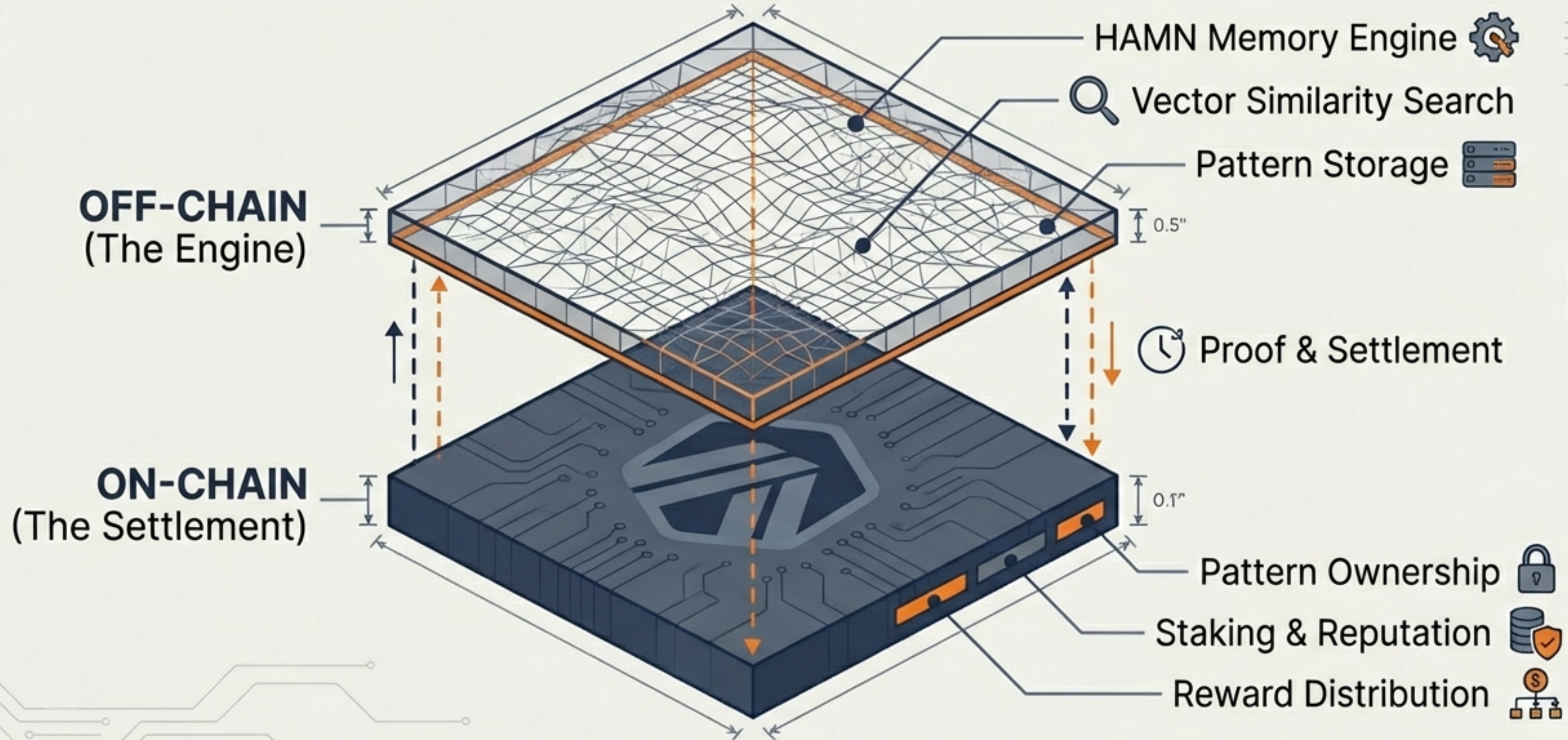
Benchmark: Agent **Auto-Strategy Builder** (Maximize Yield)



How It Works: The Associative Loop

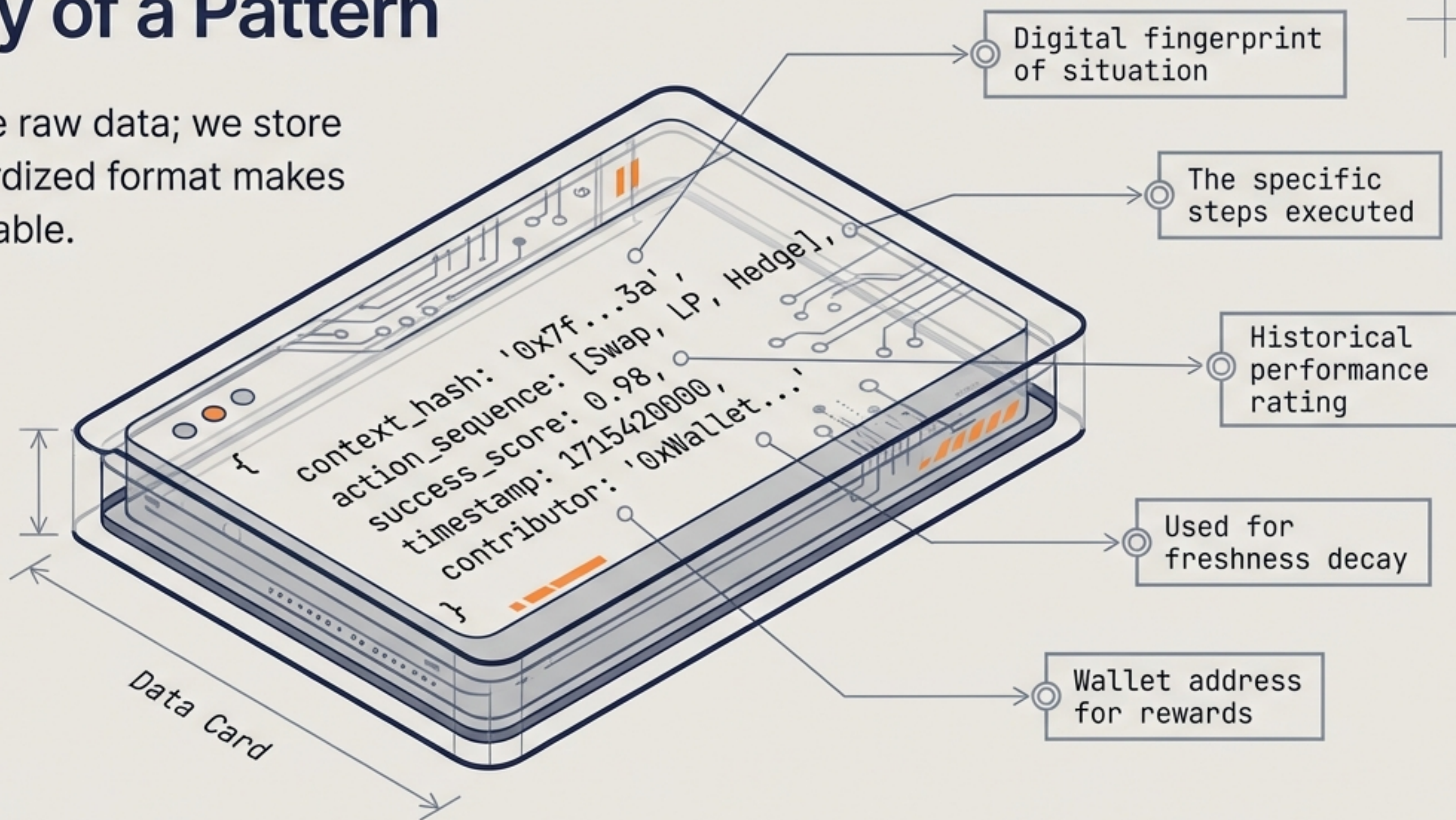


Hybrid Architecture: Speed & Trust



Anatomy of a Pattern

We do not store raw data; we store logic. A standardized format makes memory actionable.



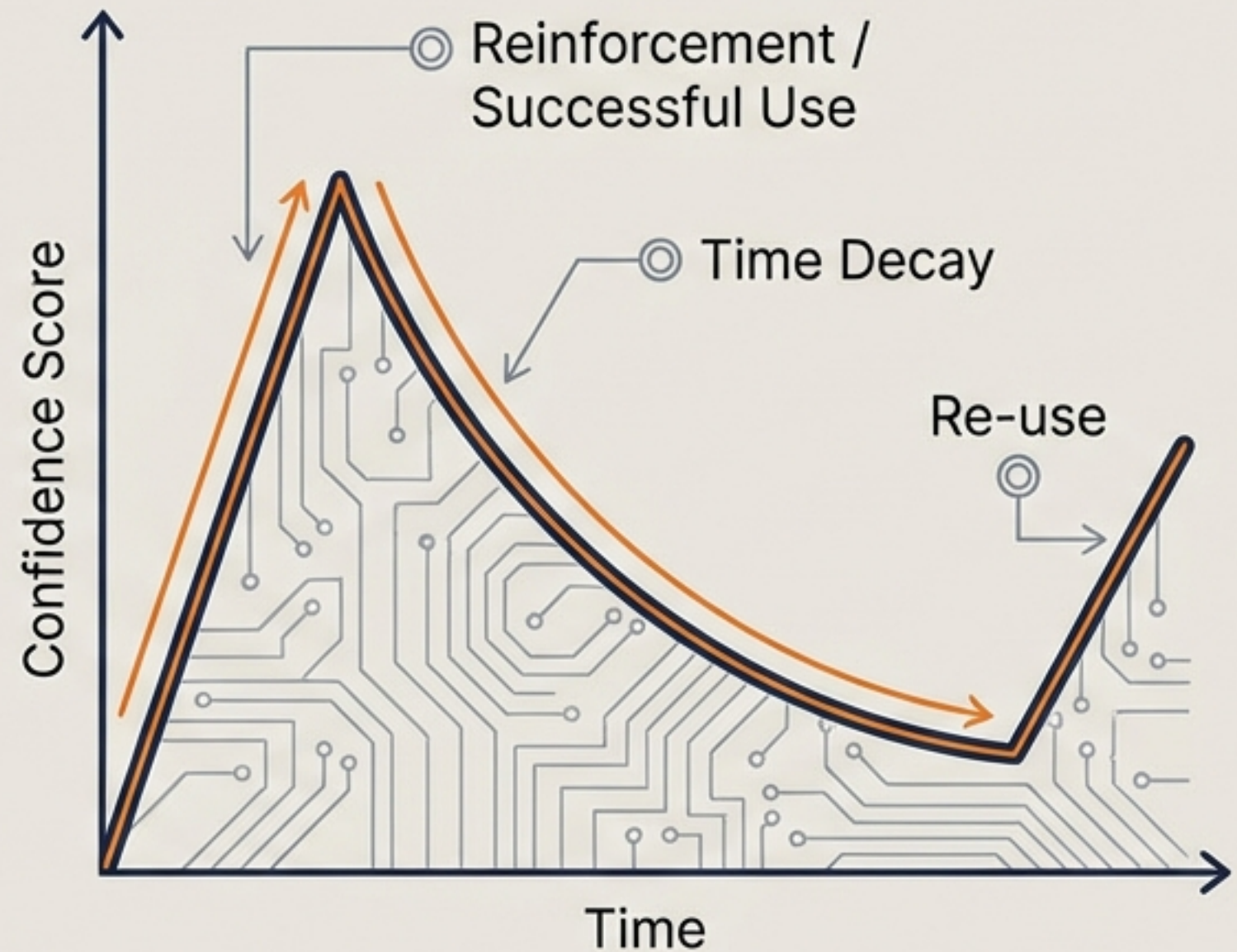
The Math of Relevance

Selection Formula:

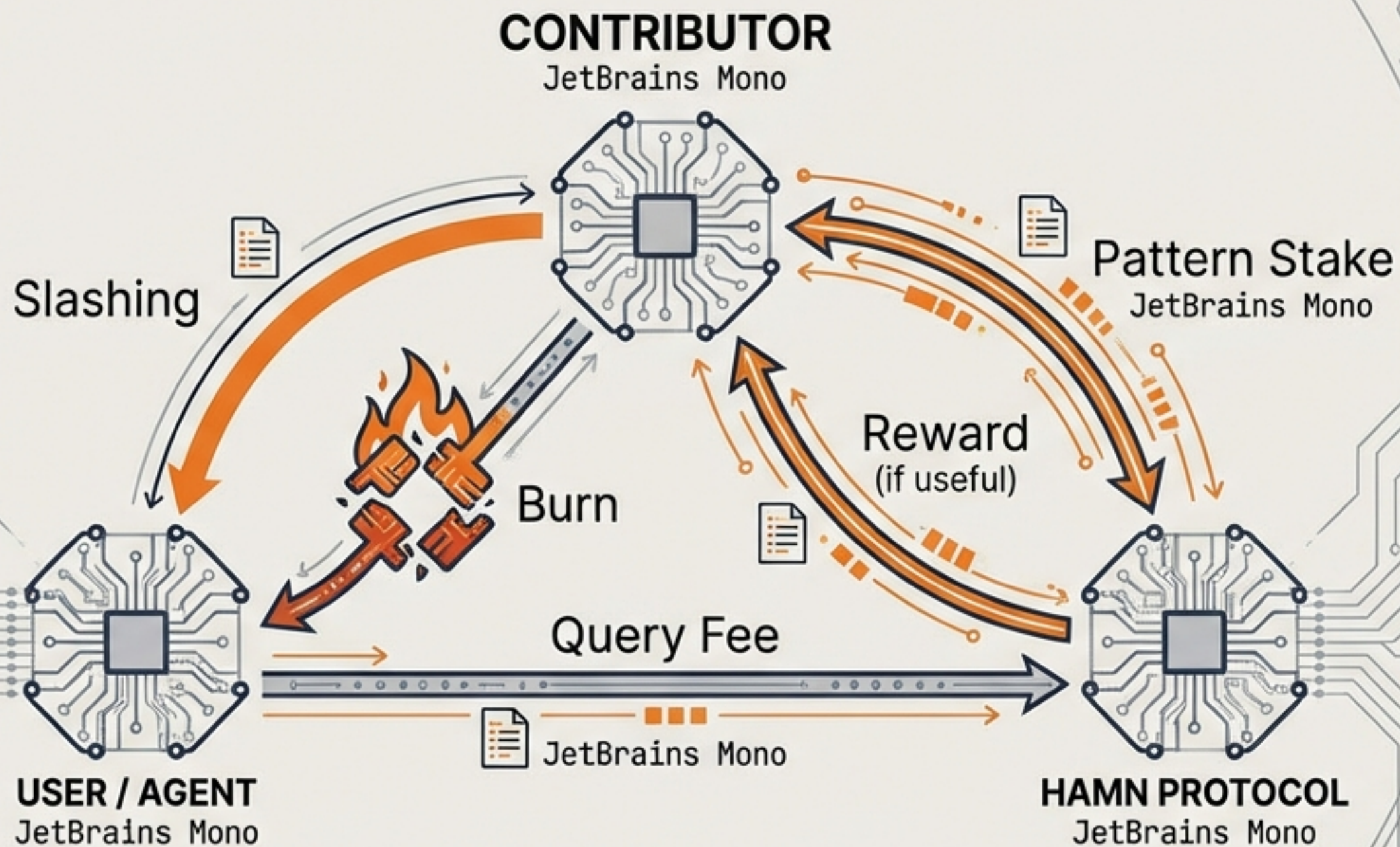
$$\text{argmax}(\text{similarity} \times \text{confidence} \times \text{freshness})$$

The Decay Logic:

$$\text{confidence} \ast= \exp(-\lambda \ast \text{time})$$

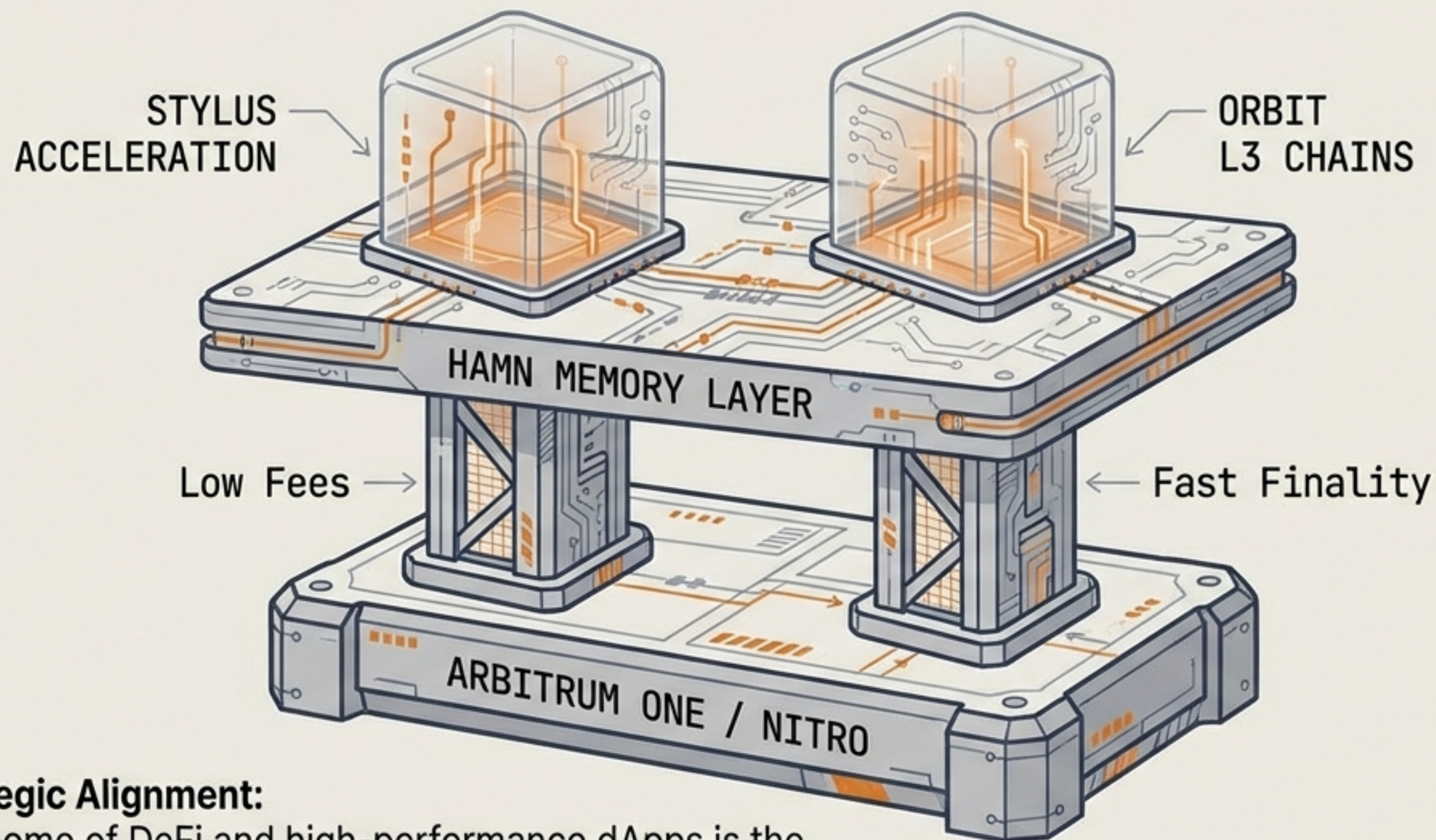


The Economy of Intelligence



$$\text{reward} \propto \text{usefulness} \times \text{reuse_count}$$

Built for the Arbitrum Ecosystem

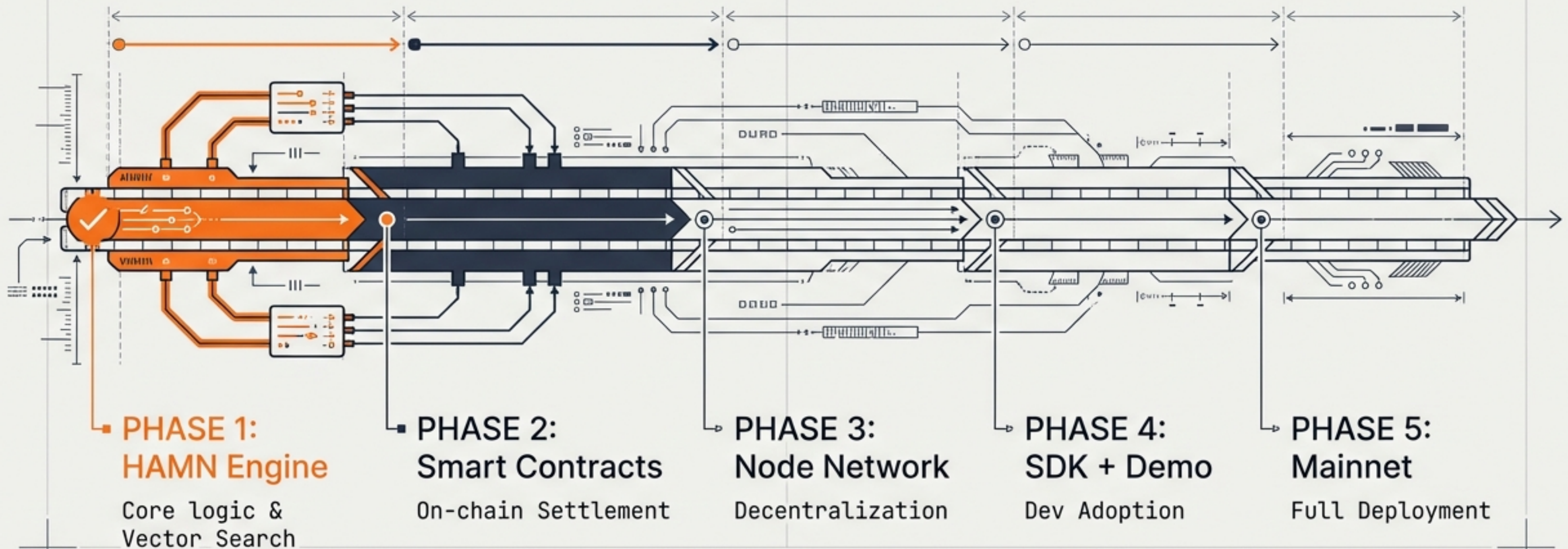


Strategic Alignment:

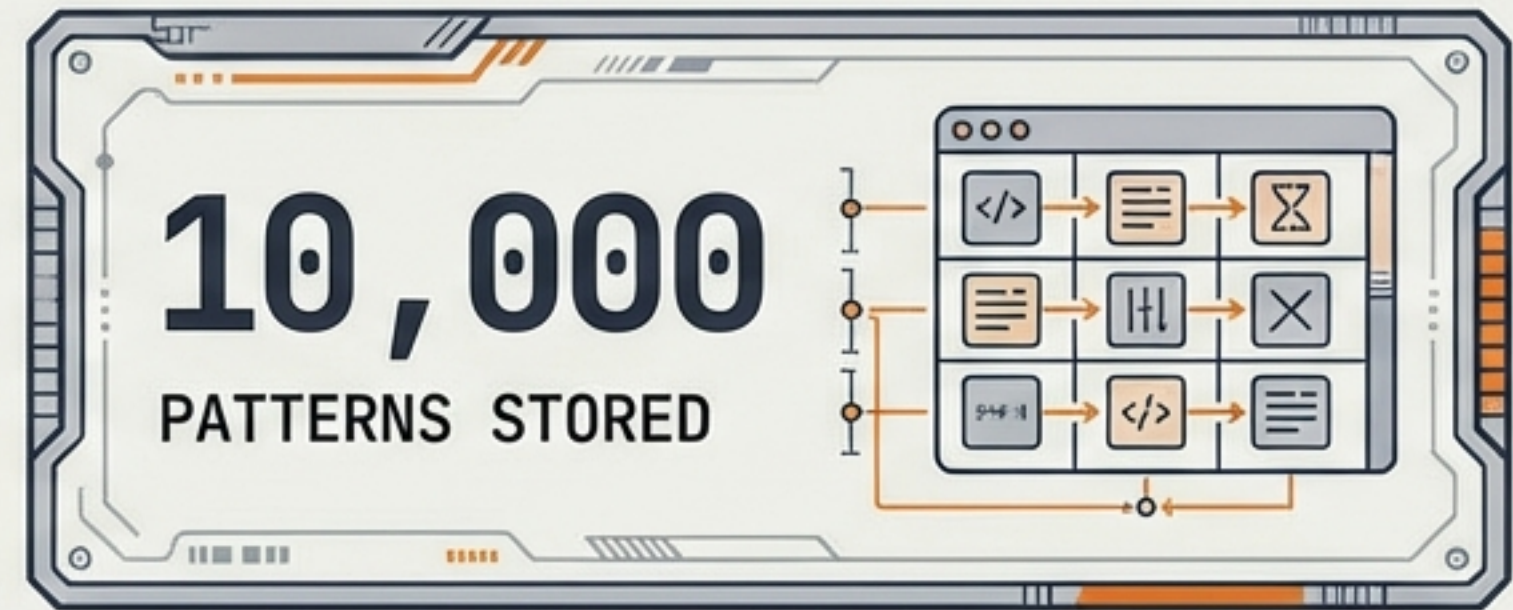
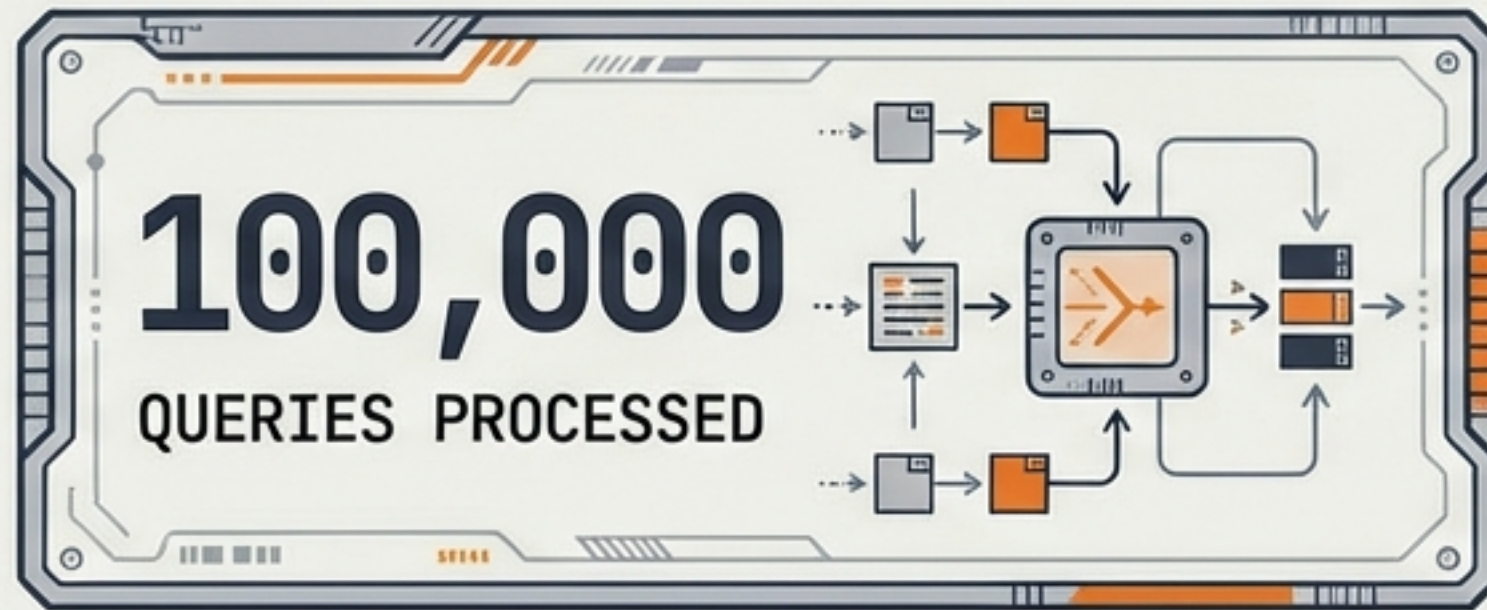
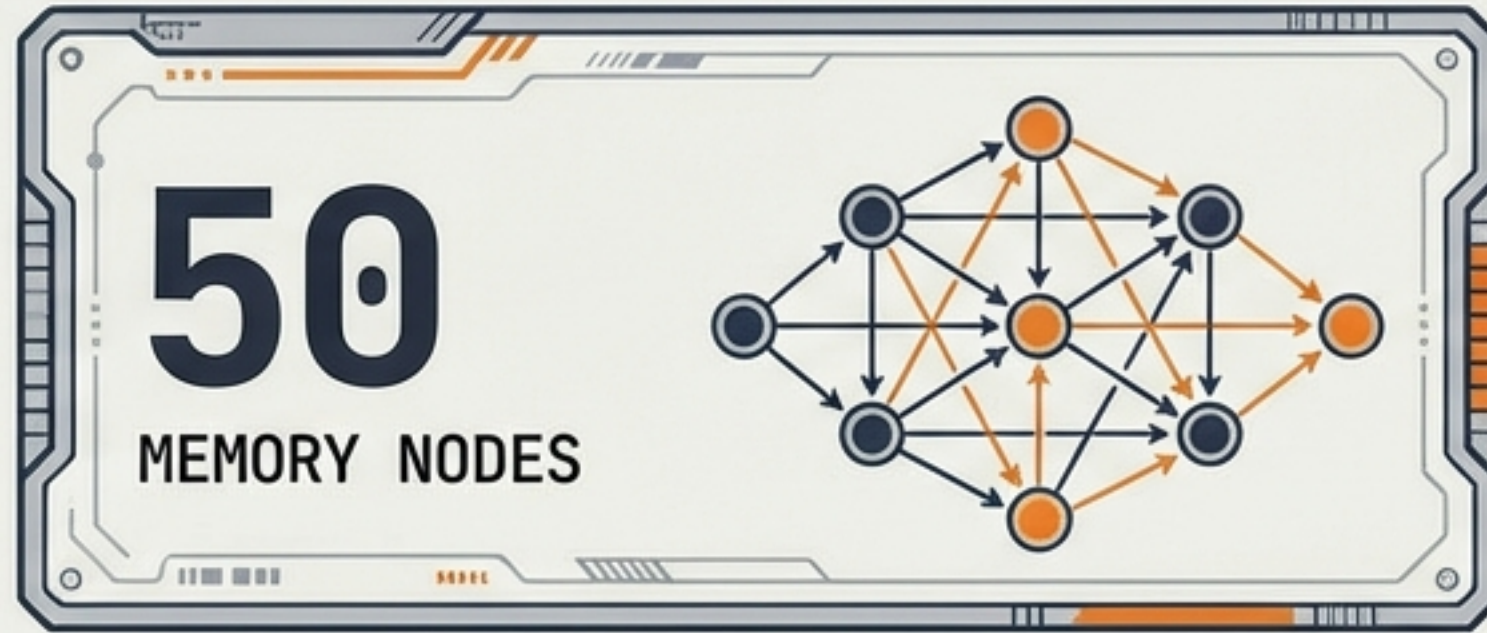
The home of DeFi and high-performance dApps is the only logical foundation for high-performance memory.

ENGINEERING BLUEPRINT | REV. A.09

Implementation Roadmap



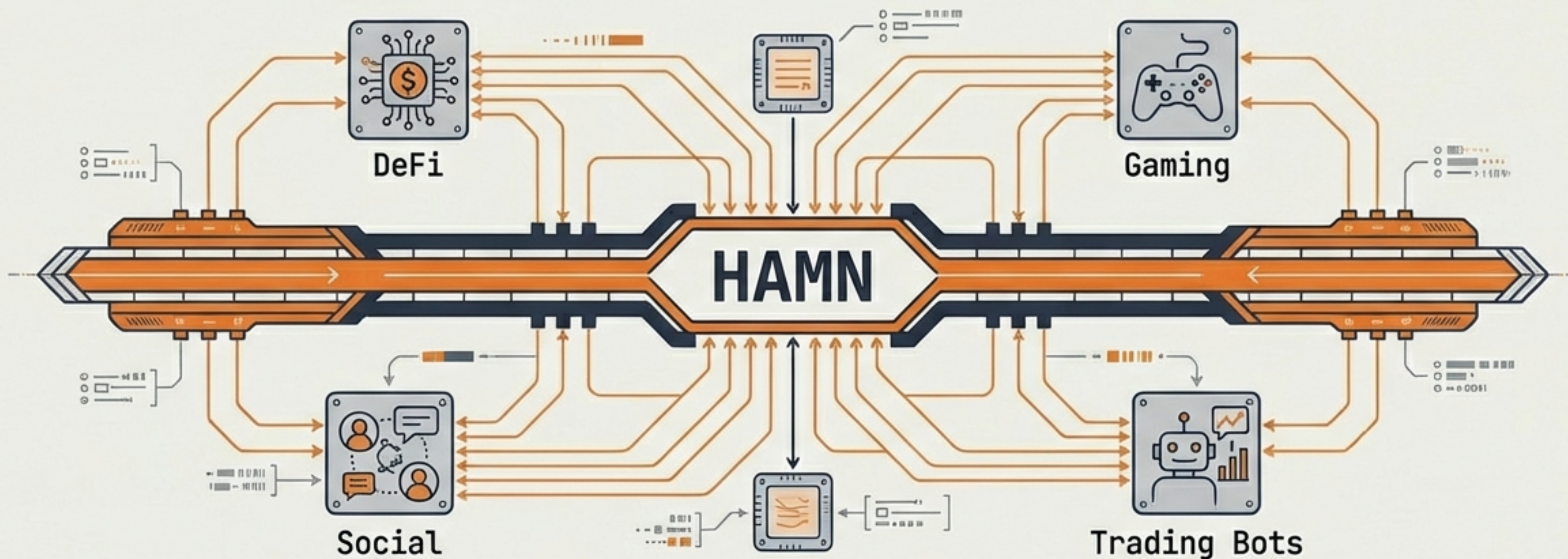
6-Month Impact Targets



Success is defined by network density and query volume.

Infrastructure for Collective Memory

HAMN is not an AI model. It is the missing link that allows all AI models to learn from each other. We are building the knowledge infrastructure for the next generation of Web3.



The transition from isolated intelligence to shared mastery begins on Arbitrum.